

Question number	Answer	Additional guidance	Mark
2(a)	An answer that includes three of the following points, with at least one similarity and one difference : Similarities • both have (circular) DNA (1) • both have ribosomes (1) • both have double membranes (1) Differences • chloroplasts have stroma and mitochondria have a matrix (1) • chloroplasts have {thylakoids / thylakoid membranes / grana / intergranal lamellae} and mitochondria have {a folded inner membrane / cristae} (1) • chloroplasts have starch grains and mitochondria do not (1)	DO NOT PIECE TOGETHER ACCEPT both have an envelope	(3)

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2(b)(i)	An answer that includes the following points: • (D) chlorophyll / light-absorbing pigments / photosystems / PSI / PSII (1) • (E) electron carriers / electron transport proteins / cytochromes (1) • (F) {ADP / adenosine diphosphate} and (G) {phosphate (ions) / Pi / PO₄²⁻} and (H) ATP / adenosine triphosphate (1)	ACCEPT named pigments ACCEPT F & G the other way around DO NOT ACCEPT P or incorrect formula for phosphate ions	(3)

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2(b)(ii)	• reduced NADP / NADPH	ACCEPT {r / red} for reduced reduced NADPH DO NOT ACCEPT NADP /reduced NAD/ NADH / reduced NADH	(1)

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2(b)(iii)	A description that includes two of the following points: • to split water (1) • electrons replace those lost by {photosystems / PSI / PSII / chlorophyll} (1) • {hydrogen ions / H⁺ / protons} involved in formation of NADPH (1)		(2)